

AION Development Commands Runbook

Clone the repository, work on the team branches, run both apps, and verify the project locally.

Repository: github.com/CodeGeniusDev/AION | Default local folder: AION

Quick start

Use the commands below in a terminal. The commands assume macOS/Linux and a shell such as zsh or bash.

1. Clone the repository

```
git clone https://github.com/CodeGeniusDev/AION.git
cd AION
git fetch origin
git branch -a
```

Team branch workflow

Use one branch per contributor. Create the branch once, then push it to GitHub. If the branch already exists, use the existing-branch commands instead.

2A. Mahnoor and Abdullah - create and publish branches

```
# Mahnoor
git switch -c mahnoor
git push -u origin mahnoor

# Abdullah
git switch -c abdullah
git push -u origin abdullah
```

2B. If the branch already exists on GitHub

```
git fetch origin
git switch mahnoor # or: git switch abdullah
git pull --ff-only origin mahnoor # or use abdullah
```

Before starting work, confirm your branch with: `git branch --show-current`

Run the application

Keep the frontend and backend in separate terminal windows so both services can run at the same time.

3. Frontend - terminal window 1

```
cd /AION/AION/apps/web
npm install
npm run dev
```

Frontend URL

<http://localhost:3000>

4. Backend - terminal window 2

```
cd /AION/AION
source .venv/bin/activate
cd apps/api
uvicorn main:app --reload --port 8000
```

Backend URLs

<http://localhost:8000> | API docs: <http://localhost:8000/docs>

If the virtual environment does not exist yet, create it from the repository root with: `python3 -m venv .venv`

Daily Git workflow

Run this sequence before and after making changes on your own branch.

5. Pull, work, commit, and push

```
git switch mahnoor # use abdullah on Abdullah's branch
git pull --ff-only origin mahnoor # use abdullah when appropriate

# make your changes, then inspect them
git status
git diff

git add .
git commit -m "Describe the change"
git push
```

Testing and verification

Run the backend tests first, then validate the frontend with linting, TypeScript checking, and a production build.

6. Backend tests

```
cd /AION/AION/apps/api
../../.venv/bin/python -m pytest -q
```

7. Frontend checks

```
cd /AION/AION/apps/web
npm run lint
npm run typecheck
npm run build
```

Useful checks

Check	Command	Expected result
Current branch	git branch --show-current	mahnoor or abdullah
Working tree	git status	Review changes before committing
Frontend	http://localhost:3000	AION web app loads
Backend	curl http://localhost:8000/health	Healthy API response

Safe collaboration notes

Do not commit API keys or local environment files. Pull with --ff-only when possible. Push only your own branch, then open a pull request when the team is ready to review the work.

AION - Artificial Intelligence Operating Nervous System